

1. Original Problem

$$\min_{x,v} f(v) \text{ s.t. } v = B'x + v_0, Ax = d, x \geq 0$$

B : path-link incidence matrix ($n \times m$); A : OD-path incidence matrix ($\ell \times n$)
 $n + m$ variables

2. Nominal Solution

Compute a reference equilibrium to obtain nominal flows

3. Path Classification (threshold τ)

Major paths (s paths): 1) Each OD has at least one major path, 2) The remaining major paths have flow larger than τ
Minor: remaining $n-s$ paths

4. Partition into Major and Minor Blocks

y : major-path flows

w : minor-path flows

B_1 ($s \times m$): major rows of B

B_2 $[(n-s) \times m]$: minor rows of B

A_1 ($\ell \times s$): major columns of A

A_2 $[\ell \times (n-s)]$: minor columns of A

5. Truncated SVD of B_2

$$B_2 \approx U_r \Sigma_r V_r'; \quad w \approx U_r z, \quad z \in \mathbb{R}^r$$
$$D = B_2' U_r \quad (m \times r); \quad M = A_2 U_r \quad (\ell \times r)$$

6. Compressed Problem

$$\min_{y,z} \hat{f}(y,z) \text{ s.t. } A_1 y + M z = d, \quad y \geq 0, \quad U_r z \geq 0$$
$$\hat{f}(y,z) = f(B_1' y + D z + v_0)$$

$s + r$ variables (much smaller than $n + m$)